

INTAKE-ARBITER -- FREE-COOLING DECISION REPORT

An agent that decides, hour by hour, whether outside air can cool a data centre.

THE SITE

Location Aligned Energy Data Center, AZ -- station KDVT, America/Phoenix
Building OSM 155739337
Facade gap Aligned Energy Data Center
Weather record not applicable -- one building, no second facade
Weather record 43,125 real hourly records from KDVT
Plume physics NOT MODELLED. No other tagged data centre lies inside the solver's validated range, so there is no neighbour intake for a plume to arrive at: the quantity does not exist here rather than being unmeasured. This is a statement about the model's domain, not a claim that recirculation here is zero. A building's own exhaust re-entering its own intake is not modelled at any site in this project.

FortyGuard data: weather and geometry are this site's own; the four measured FortyGuard level offsets and the N-26 coverage record are Ashburn's, because only Ashburn has forecast/outcome day pairs. Quote the hours for this site; quote the coverage for Ashburn.

THIS REPORT IS A SNAPSHOT, NOT THE LIVE PAGE

The interface recomputes the decision for whatever configuration a reader selects. This file was generated at build time for the ONE configuration named below, chosen because it is the configuration in which this site's agent does the most while still changing mode at least once. If the numbers on screen differ from the numbers here, the configuration differs -- compare the two lists before concluding anything else.

Day 2023-12-17 (crossing)
Plant limit 18.0 C
Notice required 3 h
Forecast skill 0.50 relative to persistence
Level anchor sensor
Condenser bank longest facade
Switch budget 1 mode changes per day
Minimum dwell 3 h
Max dew point 15.0 C (Green Grid WP#46 p.6)

WHAT THE AGENT DID ON THIS DAY

The agent ran free cooling for 3 of 24 hours with 1 mode change. The reactive incumbent took 7 hours -- 4 more -- but declared 0 unsafe hours against the agent's 0. The incumbent had to break its own switch budget 1 time to stay safe; the agent never did. 8 hours were safe but still ran chillers, because the schedule could not afford them.

Agent 3 of 24 hours free cooling, 1 mode change(s), 0 unsafe hour(s)
Incumbent 7 of 24 hours free cooling, 2 mode change(s), 0 unsafe hour(s)
the incumbent broke its own switch budget 1 time(s) to stay safe
8 hour(s) passed every gate and still ran chillers, because the schedule could not afford them

EVERY HOUR, AND THE REASON FOR IT

hh	mode	safe	binding	bound	limit	actual	margin
00	mechanical	yes	switch budget	12.220	18.0	12.220	1.179
01	mechanical	yes	switch budget	12.220	18.0	12.220	1.071
02	mechanical	yes	switch budget	11.945	18.0	11.670	0.983
03	mechanical	yes	switch budget	13.330	18.0	14.440	0.885
04	mechanical	yes	switch budget	13.055	18.0	13.890	0.810
05	mechanical	yes	switch budget	13.060	18.0	13.890	0.914

06	mechanical	yes	switch budget	14.720	18.0	12.780	1.157
07	mechanical	yes	switch budget	16.390	18.0	13.330	1.772
08	mechanical	no	dry-bulb	19.445	18.0	17.220	1.692
09	mechanical	no	dry-bulb	21.670	18.0	22.220	1.414
10	mechanical	no	dry-bulb	27.495	18.0	33.330	1.466
11	mechanical	no	dry-bulb	28.330	18.0	32.780	1.084
12	mechanical	no	dry-bulb	24.995	18.0	22.220	0.892
13	mechanical	no	dry-bulb	35.280	18.0	32.780	0.689
14	mechanical	no	dry-bulb	35.560	18.0	35.000	0.515
15	mechanical	no	dry-bulb	31.390	18.0	37.780	0.649
16	mechanical	no	dry-bulb	34.445	18.0	34.440	0.634
17	mechanical	no	dry-bulb	28.615	18.0	21.670	0.804
18	mechanical	no	dry-bulb	28.330	18.0	19.440	1.069
19	mechanical	no	dry-bulb	25.550	18.0	18.330	1.113
20	mechanical	no	dry-bulb	19.170	18.0	18.890	0.964
21	FREE	yes	-	17.780	18.0	17.780	1.068
22	FREE	yes	-	17.220	18.0	16.670	1.325
23	FREE	yes	-	17.505	18.0	16.670	1.103

bound = the agent's 90 %-nominal upper bound on intake air, forecast plus margin
actual = what the intake temperature turned out to be, from the station record
margin = group-conditional forecast error for that hour of day, plus plume spread

THE REASONING, HOUR BY HOUR

00:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 12.220 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

01:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 12.220 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

02:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 11.945 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

03:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 13.330 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

04:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 13.055 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is

also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

05:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 13.060 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

06:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 14.720 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

07:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 16.390 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

08:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 19.445 C against a 18.0 C limit, so it fails by 1.445 C. It would take a limit 1.445 C higher, or a bound 1.445 C tighter, to change this hour.

09:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 21.670 C against a 18.0 C limit, so it fails by 3.670 C. It would take a limit 3.670 C higher, or a bound 3.670 C tighter, to change this hour.

10:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 27.495 C against a 18.0 C limit, so it fails by 9.495 C. It would take a limit 9.495 C higher, or a bound 9.495 C tighter, to change this hour.

11:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 28.330 C against a 18.0 C limit, so it fails by 10.330 C. It would take a limit 10.330 C higher, or a bound 10.330 C tighter, to change this hour.

12:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 24.995 C against a 18.0 C limit, so it fails by 6.995 C. It would take a limit 6.995 C higher, or a bound 6.995 C tighter, to change this hour.

13:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 35.280 C against a 18.0 C limit, so it fails by 17.280 C. It would take a limit 17.280 C higher, or a bound 17.280 C tighter, to change this hour.

14:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 35.560 C against a 18.0 C limit, so it fails by 17.560 C. It would take a limit 17.560 C higher, or a bound 17.560 C tighter, to change this hour.

15:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 31.390 C against a 18.0 C limit, so it fails by 13.390 C. It would take a limit 13.390 C higher, or a bound 13.390 C tighter, to change this hour.

16:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 34.445 C against a 18.0 C limit, so it fails by 16.445 C. It would take a limit 16.445 C higher, or a bound 16.445 C tighter, to change this hour.

17:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 28.615 C against a 18.0 C limit, so it fails by 10.615 C. It would take a limit 10.615 C higher, or a bound 10.615 C tighter, to change this hour.

18:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 28.330 C against a 18.0 C limit, so it fails by 10.330 C. It would take a limit 10.330 C higher, or a bound 10.330 C tighter, to change this hour.

19:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 25.550 C against a 18.0 C limit, so it fails by 7.550 C. It would take a limit 7.550 C higher, or a bound 7.550 C tighter, to change this hour.

20:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 19.170 C against a 18.0 C limit, so it fails by 1.170 C. It would take a limit 1.170 C higher, or a bound 1.170 C tighter, to change this hour.

21:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 17.780 C, which is 0.220 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.068 C, and the margin is measured, not chosen: 1.068 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

22:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 17.220 C, which is 0.780 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.325 C, and the margin is measured, not chosen: 1.325 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

23:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 17.505 C, which is 0.495 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.103 C, and the margin is measured, not chosen: 1.103 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

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